

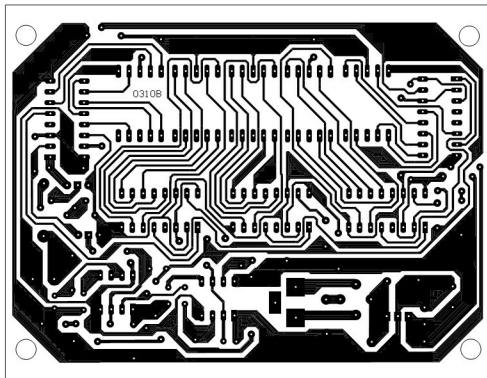


Fig. 3: A single-side, actual-size PCB layout for frequency counter

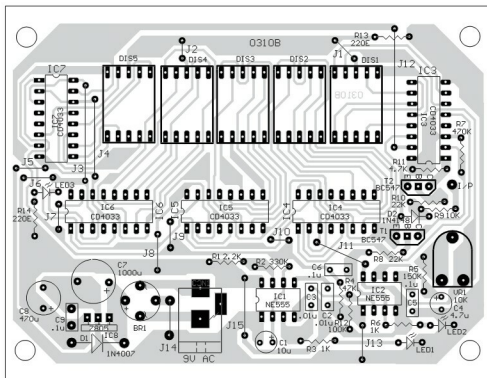


Fig. 4: Component layout for the PCB

PARTS LIST

Semiconductors:

- IC1, IC2 - NE555 timer
- IC3-IC7 - CD4033 decade counter/driver
- IC8 - 7805, 5V regulator
- T1, T2 - BC547 npn transistor
- D1 - 1N4007 rectifier diode
- D2 - 1N4148 switching diode
- LED1-LED3 - 5mm LED
- D1S1-D1S5 - LTS543 common-cathode 7-segment display
- BR1 - 1A bridge rectifier

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1 - 2.2-kilo-ohm
- R2 - 330-kilo-ohm
- R3, R6 - 1-kilo-ohm
- R4 - 47-kilo-ohm
- R5 - 150-kilo-ohm
- R7 - 470-kilo-ohm
- R8, R10 - 22-kilo-ohm
- R9 - 10-kilo-ohm
- R11 - 4.7-kilo-ohm
- R12 - 100-kilo-ohm
- R13, R14 - 220-ohm
- VR1 - 100-kilo-ohm trimpot

Capacitors:

- C1 - 10 μ F, 16V electrolytic
- C2, C3, C5 - 0.01 μ F ceramic disk
- C4 - 4.7 μ F, 16V electrolytic
- C6, C9 - 0.1 μ F ceramic disk
- C7 - 1000 μ F, 25V electrolytic
- C8 - 470 μ F, 16V electrolytic

Miscellaneous:

- X1 - 230V AC primary to 9V, 500mA secondary transformer

it is set precisely to one second.

The monostable multivibrator is triggered by the astable multivibrator built around another IC NE555 (IC1). The time period of the astable multivibrator is determined by the combination of resistors R1 and R2 and capacitor C1. Here IC1 is designed to output a square wave having two-second 'high' and two-second 'low' periods.

The monostable multivibrator (IC2) is triggered whenever the output of the astable multivibrator (IC1) goes low. The output of IC2 instantly goes

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